Answer Key — Mixtures

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

Q1 **(b) keep their own properties** — a mixture's components keep their own properties; no reaction occurs.

Q2 **(c) sugar dissolved in water** — a sugar solution is evenly mixed — uniform.

Q3 **(b) copper and zinc** — brass = copper + zinc (bronze = copper + tin).

Q4 **(c) carbon dioxide** — CO_2 forms calcium carbonate, turning lime water milky.

Q5 **(b) False** — air is a uniform mixture, not a compound.

Q6 **alloy** — such metal mixtures are alloys.

Q7 **components** — the parts of a mixture are its components.

Short answer

Q8 A combination of two or more substances that keep their own properties; uniform: sugar solution; non-uniform: sand in water (or salad).

Q9 Its gases are evenly mixed and cannot be seen separately; mainly nitrogen (~78%), oxygen, argon, carbon dioxide and water vapour.

Q10 Stainless steel: iron, nickel, chromium, carbon; brass: copper + zinc; bronze: copper + tin.

Long / application

Q11 Uniform mixtures have evenly distributed components that can't be seen separately (sugar solution, air); non-uniform have visible/uneven components (salad, sand in water). Alloys (steel, brass, bronze) are uniform mixtures of metals — they look the same throughout.

Q12 Leave colourless lime water (calcium hydroxide) exposed to air; it turns milky. Word equation: Calcium hydroxide + Carbon dioxide → Calcium carbonate + Water. The milky calcium carbonate shows CO_2 is present in air.